

In a generic case we could have $A(z)$ not monic.

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$$\tilde{A}(z) = \tilde{a}_0 + \tilde{a}_1 z^{-1} + \dots + \tilde{a}_m z^{-m}$$

and correspondingly

$$\tilde{B}(z) = \tilde{b}_0 + \tilde{b}_1 z^{-1} + \dots + \tilde{b}_m z^{-m}$$

$\tilde{A}(z), \tilde{B}(z)$
co-prime

Let consider the transf. function

$$G(z) = \frac{\tilde{B}(z)}{\tilde{A}(z)}$$

Note that in order to ensure $G(z)$ is a proper transf. function $\Rightarrow \tilde{a}_0 \neq 0$

$\Rightarrow$ we can divide $\tilde{B}(z), \tilde{A}(z)$ by $\tilde{a}_0$

$$\Rightarrow G(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_m z^{-m}}{1 + a_1 z^{-1} + \dots + a_m z^{-m}}$$

$$\text{where } b_i = \frac{\tilde{b}_i}{\tilde{a}_0} \quad i = 0, \dots, m$$

$$a_i = \frac{\tilde{a}_i}{\tilde{a}_0} \quad i = 1, \dots, m$$

$\Rightarrow$ Assuming $a_0 = 1$ is not a limitation.

Let now consider the term

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$$H(z)u(k) = \frac{D(z)}{C(z)} u(k)$$

for generic polynomials $\tilde{D}(z), \tilde{C}(z)$ not necessarily monic.

$$\tilde{D}(z) = \tilde{d}_0 + \tilde{d}_1 z^{-1} + \dots + \tilde{d}_q z^{-q}$$

$$\tilde{u}(k) \sim WN(0, \tilde{\lambda}^2)$$

$$\tilde{C}(z) = \tilde{c}_0 + \tilde{c}_1 z^{-1} + \dots + \tilde{c}_l z^{-l}$$

$\tilde{c}_0 \neq 0$ for having $H(z) = \frac{\tilde{D}(z)}{\tilde{C}(z)}$ a proper transf. function

$$\tilde{D}(z) = \frac{1}{\tilde{c}_0} \tilde{D}(z) = \tilde{d}_0 + \tilde{d}_1 z^{-1} + \dots + \tilde{d}_q z^{-q}$$

$$\text{where } \tilde{d}_i = \frac{d_i}{\tilde{c}_0}$$

$$C(z) = \frac{1}{\tilde{c}_0} \tilde{C}(z) = 1 + c_1 z^{-1} + \dots + c_l z^{-l}$$

$$c_i = \frac{\tilde{c}_i}{\tilde{c}_0}$$

$$\tilde{H}(z) = \frac{\tilde{d}_0 + \tilde{d}_1 z^{-1} + \dots + \tilde{d}_q z^{-q}}{1 + c_1 z^{-1} + \dots + c_l z^{-l}} \tilde{u}(k) =$$

if $\tilde{d}_0 \neq 0$

$$= \frac{\tilde{d}_0}{\tilde{c}_0} \frac{1 + d_1 z^{-1} + \dots + d_q z^{-q}}{C(z)} \underbrace{\tilde{d}_0 \tilde{u}(k)}_{\tilde{u}(k)}$$

$$d_i = \frac{\tilde{d}_i}{\tilde{d}_0}$$

NOTE that $\tilde{d}_0 \tilde{u}(k)$ is still a white noise with zero mean and ...

... variance = ?

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$$\text{Var}(\tilde{d}_0 \tilde{m}(k)) = E\left\{\left(\tilde{d}_0 \tilde{m}(k)\right)^2\right\} = \tilde{d}_0^2 E\left\{\tilde{m}^2(k)\right\} = \tilde{d}_0^2 \tilde{\gamma}_2$$

$$\tilde{H}(z) \cdot \tilde{m}(k) = \tilde{H}(z) \tilde{m}(k)$$

$$\text{with } \tilde{H}(z) = \frac{D(z)}{C(z)} \quad C(z), D(z) \text{ monic}$$

if $\tilde{d}_0 = 0$

$$\tilde{H}(z) \tilde{m}(k) = \frac{\tilde{d}_1 z^{-1} + \tilde{d}_2 z^{-2} + \dots + \tilde{d}_q z^{-q}}{C(z)} \tilde{m}(k) =$$

if $\tilde{d}_1 \neq 0$

$$= \frac{1 + \tilde{d}_1 z^{-1} + \dots + \tilde{d}_q z^{-q}}{C(z)} \boxed{\tilde{d}_1 z^{-1} \tilde{m}(k)}$$

$$\tilde{d}_i = \frac{\tilde{d}_{i+1}}{\tilde{d}_1}$$

$$q = \tilde{q} - 1$$

$z^{-1}(\tilde{d}_1 \tilde{m}(k))$ is still a WN with zero mean and variance $\tilde{d}_1^2 \tilde{\gamma}_2$ because it is $\tilde{m}(k)$ delayed of one step!

$$\tilde{m}(k) \sim \text{WN}(0, \tilde{d}_1^2 \tilde{\gamma}_2)$$

As $\tilde{m}(k)$ was white shifting maintains whiteness.

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$\Rightarrow$ we can "shift" until we find a coefficient $\tilde{d}_i \neq 0$ and shift the noise of i -steps.

$\Rightarrow$ without loss of generality $A(z), C(z), D(z)$ can be considered monic.

NOTES

- $\tilde{m}(k) \sim WN(0, \tilde{\lambda}^2)$
- $n(k) \sim WN(0, \underbrace{\tilde{d}_i^2 \tilde{\lambda}^2}_{\lambda^2})$
- $\lambda^2 = \left(\frac{\tilde{d}_i}{C_0}\right)^2 \tilde{\lambda}^2$

the mean still is zero
but the variance has changed.

We will not use λ^2 in the identification process.

•) Why $B(z)$ cannot be considered monic?

Compare $G(z)u(k) = \frac{B(z)}{A(z)}u(k)$ and $\frac{D(z)}{C(z)}n(k)$

We cannot divide by b_0 the polynomial $B(z)$ and write

$$\frac{1 + \frac{b_1}{b_0} z^{-1} + \dots + \frac{b_m}{b_0} z^{-m}}{A(z)} \quad \underbrace{b_0 u(k)}_{\text{Eq. 1.48}}$$

two problems. •) b_0 will be unknown in our identification process. $u(k)$ is measured and used for identification $\Rightarrow$ NOT viable solution

•) b_0 could be zero!

In many situations reasonably there is some delay between in & out.

In this case $b_0 = 0$ [in some books

Also $b_1, b_2, \dots$ could be zero. $\left. \begin{array}{l} \text{ } \\ \text{ } \end{array} \right\} \begin{array}{l} b_0 \text{ is assumed to} \\ \text{be zero} \end{array}$

$\Rightarrow B(z)$ cannot be considered monic in general.

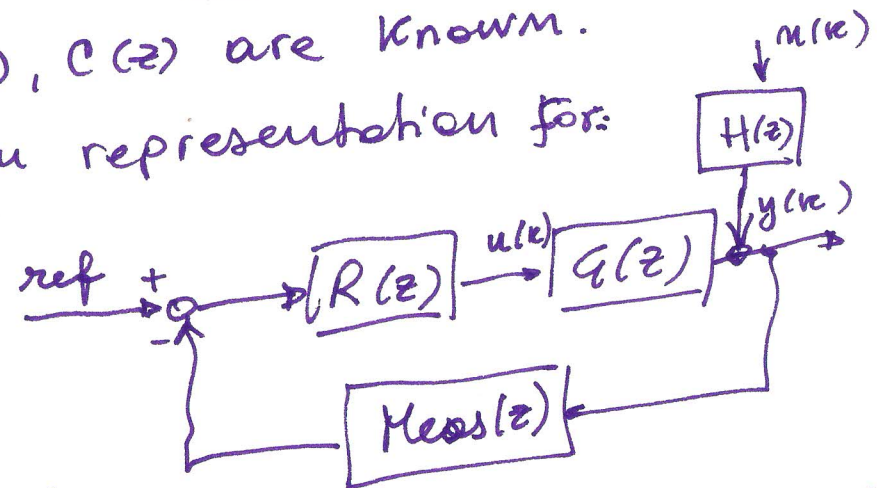
Let consider our system

$$y(k) = \underbrace{\frac{B(z)}{A(z)}}_{=G(z)} u(k) + \underbrace{\frac{D(z)}{C(z)}}_{=H(z)} n(k)$$

and assume $B(z)$, $A(z)$, $D(z)$, $C(z)$ are known.

We can use the system representation for:

- Control.



closed-loop scheme

try to design $R(z)$ ($\text{Meas}(z)$) to obtain some perform.

↓
possibly
given

- Simulation

give some control ξ in open loop or some reference in closed loop and also

give some WN sequence $n(k) \sim \text{WN}(0, \sigma^2)$

we should estimate the variance!

and see what the effect on $y(k)$

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- Prediction Assume the inputs & outputs of the system have been measured until instant k and assume we are going to apply a new input $u(k+1)$ (known).
What the output $y(k+1)$ will be?

As the system is time invariant, the Prediction Problem in " k " can be reformulated in " $k-1$ ";

- Assume inputs & outputs are known until instant " $k-1$ " and assume also $u(k)$ will be applied. What $y(k)$ will be?

$$\begin{matrix} u(0), u(1) \dots u(k) \\ y(0), y(1) \dots y(k-1) \end{matrix} \Rightarrow \hat{y}(k)$$

↑
Prediction of $y(k)$
based on past values.